



LEBANESE INTERNATIONAL UNIVERSITY
MATH210: CALCULUS II (3.0 credits)
Course Detailed Syllabus
Summer 2023-2024

A. Course Description:

This is the second course in the Calculus sequence. The course material includes logarithmic, exponential, and trigonometric functions, their inverses and their derivatives, integration techniques, improper integrals, sequences, infinite series, tests of convergence, alternating series, power series, polar coordinates and its application.

B. Course Objectives:

This course aims to:

1. Provide students with an understanding of functions.
2. Compute derivatives, proper integrals and improper integrals.
3. Provide students with an understanding of series.
4. Introduce polar coordinates and its application.

C. Student's Learning Outcomes:

Upon successful completion of this course, the student should be able to:

1. Use differentiation rules to find the derivative of exponential functions, logarithmic functions and inverse trigonometric functions.
2. Apply appropriate techniques to solve integrals with rational, exponential, logarithmic and trigonometric functions.
3. Determine the nature of convergence of an Improper integral by solving the integral or by using appropriate tests.
4. Interpret the concept of a series as an infinite sum of a sequence, examine its convergence and identify the nature of convergence of some particular series.
5. Apply the polar coordinates to graph a polar curve, and calculate its length and the area enclosed by two curves.

D. Topics Covered and detailed schedule:

Week	Topic	Examples and Exercises
1/7	Revision for derivatives and integration rules from Calculus I	
2/7	7.2 Natural Logarithms (No need for logarithmic differentiation)	Examples :1,2,3,4 Exercises: 39,41, 44,45, 48 P376
3/7	7.3 The Exponential Function	Examples: 2, 3, 5, 6 Exercises:21,33, 34, 38, 43,57 P385
4/7	7.6 Inverse Trigonometric Functions	Examples: 1,4,5,6,7 Exercises: 23, 25, 33, 46, 47, 59,66, 67, 72, 79, 81, 87 P413 & 414
8/7	vacation	
9/7	8.1 Integration by Parts	Examples :1,2,3,4,6,7,8 Exercises: 10, 11, 13,35 P441,442
10/7	8.2 Trigonometric integrals	Examples: 1,2,3,4,5,6 Exercises: 6, 21, 24, 38, 42 P448
11/7	Exam 1	
15/7	8.3 Trigonometric substitution	Examples:1,2,3 Exercises: 4, 7, 9, 19, 36, 38, 43 P452
16/7	8.4 Integration of Rational functions by partial fractions	Examples: 1,2,3,4,5,6 Exercises: 12, 25, 33, 39, 41 P461, 462
17/7	Vacation	
18/7	8.7 Improper integrals	Examples:1,2,3,4,5,7,8,9 Exercises: 7, 10, 19, 29, 34, 40, 47, 51, 56 61 P487

Week	Topic	Examples and Exercises
22/7	10.1 Sequences	Examples: 3,4,5,7,8,9 Exercises: 29, 31, 45, 53, 58, 69 P542
23/7	10.2 Infinite series	Examples : 1,2,5,7,9 Exercises: 13, 27, 34, 35, 40, 45, 54 P551
23-24/7	10.4 Comparison tests	Examples : 1,2 Exercises: :9, 11, 20, 26, 29 P562
24/7	10.5 The Ratio and Root tests	Examples: 1,3 Exercises: 3, 5, 10, 11 P567
25/7	Exam2	
29/7	10.6 Alternating Series, Absolute and conditional convergence	Examples: 1, 4 Exercises: 1, 18, 35 P573
29-30/7	10.7 Power series	Examples: 1,3 Exercises: 2, 4, 16 P582
30/7	10.8 Taylor and Maclaurin's series	Examples : 2,3 Exercise: 2 p 588
31/7	11.3 Polar coordinates	Examples: 1,2,3,5,6 Exercises: 3, 27, 63 P630, 631
31/7	11.4 Graphing in polar coordinates	Example: 1 Exercises 21, 23 P635
1/8	11.5 Areas and lengths in polar coordinates	Examples : 1,2,3 Exercises: 11, 16, 28 P638 & 639

E. Textbook:

Thomas' Calculus, Twelfth Edition, Addison – Wesley Publishing Company.

F. Prerequisites–Corequisite:

Prerequisites: Math160 (Calculus I)

Corequisites: Math270 (Ordinary Differential Equation)

G. CourseAssessment:

Evaluation	Weight	Chapters Covered	CLO's
Attendance	6 %		
Exam 1	27 %	Chapters 7 and 8.1	1 (20%) and 2 (80%)
Exam 2	27%	Chapter 8	2 (80%) and 3 (20%)
Final	40 %	Chapters 10 and 11	4 (80%) and 5 (20%)